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Autonomous Networking

Networks that observe, decide, and adapt

Gaia Maselli

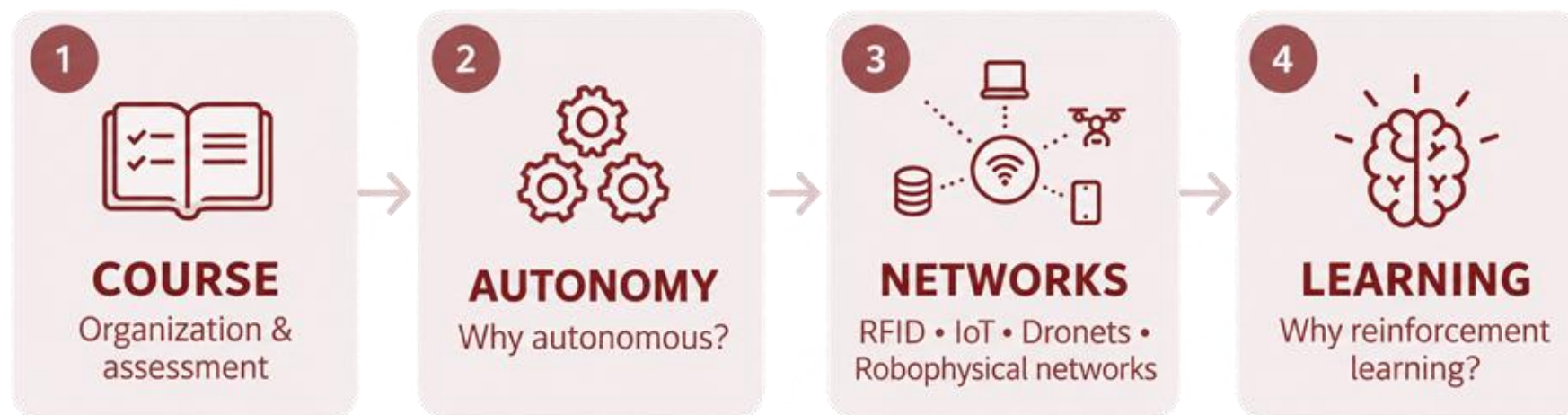
Department of Computer Science
Sapienza University of Rome

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Today's plan



From networking problems to autonomous decision making



Course logistics



CLASS SCHEDULE

Thursday 8:30–11:00

Friday 9:00–11:00

Room G50



COURSE MATERIAL

Slides (available on classroom)

Scientific papers ACM Digital Library, IEEE Xplore

Textbook

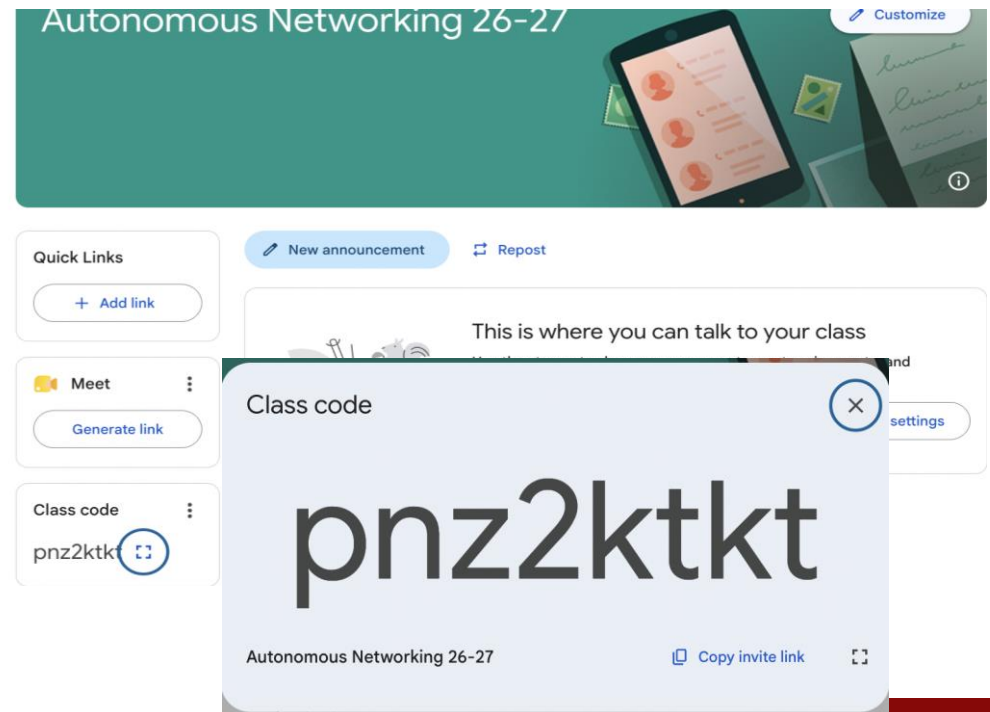
Sutton & Barto, *Reinforcement Learning: An Introduction*

Course logistics

■ Google Classroom site

- You need an institutional account
- <https://classroom.google.com/c/MjM1MzQwMDUzNzJa?cjc=nry6cprp>
- pnz2kttk

- Class info (lectures, exams)
- Material (slides, scientific papers)
- News (including delayed/canceled lectures) will be available on classroom



Assessment

- **All topics covered in the course may be assessed in the exam.**
- To pass the exam, students can choose **ONE** of the following options:

1. Written exam

During the official exam session, at the end of the course.

2. Mid-term exam + oral presentation of either:

- a research paper, or
- a project.

The oral presentation will take place before the end of the course.

Assessment: midterm + integration

- Only the students taking a grade higher than 26 can do the integration (may vary depending on the number of students)
- The presentation of the assigned scientific article will take place in class during the class hours (according to a schedule established after the mid term exam).
- Oral presentation can take place only during the course (in case you miss it you need to do the final exam)
- Oral presentation requires students to be critical (comment the presented paper)
- By the end of October, students interested in the second examination mode must inform the instructor (a google form will be provided)



Prerequisites

- Computer networks course
 - TCP/IP network stack
 - MAC and routing protocols
- Basic notions of probability

Course outline (1/3)

1 Networking systems and technologies



Wireless Sensor Networks
Sensing the physical world



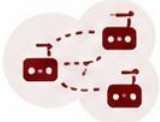
Internet of Things
Connecting devices at scale



RFID and Backscatter Networks
Communication with limited energy



Unmanned Aerial Networks (Drones)
Aerial connectivity and mobility



Robophysical Networks
Collective behavior and networked robots



Basic notions of network performance evaluation
Metrics, modeling and experimentation



NETWORKS



DECISION MAKING



AUTONOMOUS NETWORKING

Course outline (2/3)

Reinforcement Learning

1



RL problem formulation

- states / observations
- actions
- rewards

2



Core challenges

- exploration vs. exploitation
- delayed consequences

3



RL techniques

- Multi-Armed Bandits
- Q-Learning



NETWORKS



DECISION MAKING



AUTONOMOUS NETWORKING

Course outline (3/3)

Autonomous Networking

1



Learning-based approaches

- Adaptive channel access
- Adaptive routing

2



Adaptive approaches without learning

- Rule-based adaptation
- Distributed coordination



This part will be mainly explored through student presentations.



NETWORKS



DECISION MAKING



AUTONOMOUS NETWORKING

What does “autonomous” mean?

OBSERVE

The environment and its operating conditions

DECIDE

which action is appropriate

ACT

Without continuous human intervention

ADAPT

when conditions change

 **OBSERVE** →  **DECIDE** →  **ACT** →  **ADAPT**



In a network

OBSERVE

Traffic, channel, topology, energy, environment

→

DECIDE

Select an appropriate networking action

→

ACT

Change transmission, channel access, routing, mobility

→

ADAPT

React to changing conditions

Example



Suppose you have a smart home with many devices deployed in the environment



Communication channel is shared by all devices



Video streaming is taking most of bandwidth



Suddenly there is smoke in the kitchen



All devices monitoring the kitchen should be able to send their data without any delay (able to gain channel access in short time)

Example (cont.)

- The network **observes** that smoke has been detected and that the channel is congested.
- It must **decide** what traffic should be prioritized.
- It **acts** by changing the channel-access behavior.
- It **adapts** to the new operating conditions.

This is just one example. Where do we need autonomous networking?

Where do we need autonomous networking?



Which networks?

Mainly wireless

- RFID/ Backscatter networks
- Sensor & IoT
- Unmanned networks (Dronet)
- Robophysical networks



RFID, sensor networks, and IoT



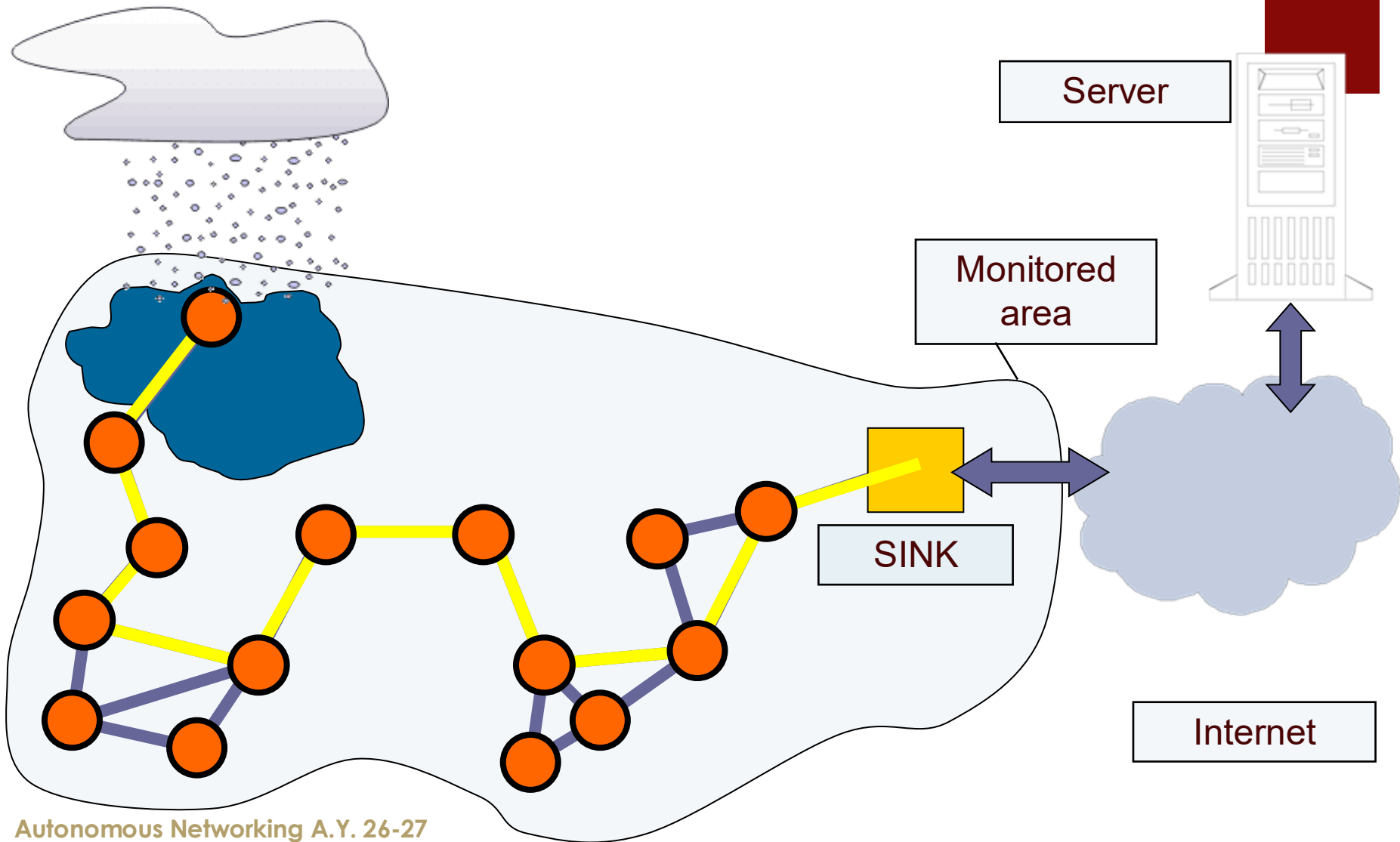
RFID

- Radio Frequency Identification



Sensor networks (sense the environment)

18



IoT network

- Smart devices
 - Sensors
 - RFID
 - Wi-Fi
- Smart environments
- Smart homes



Smart homes/Smart buildings

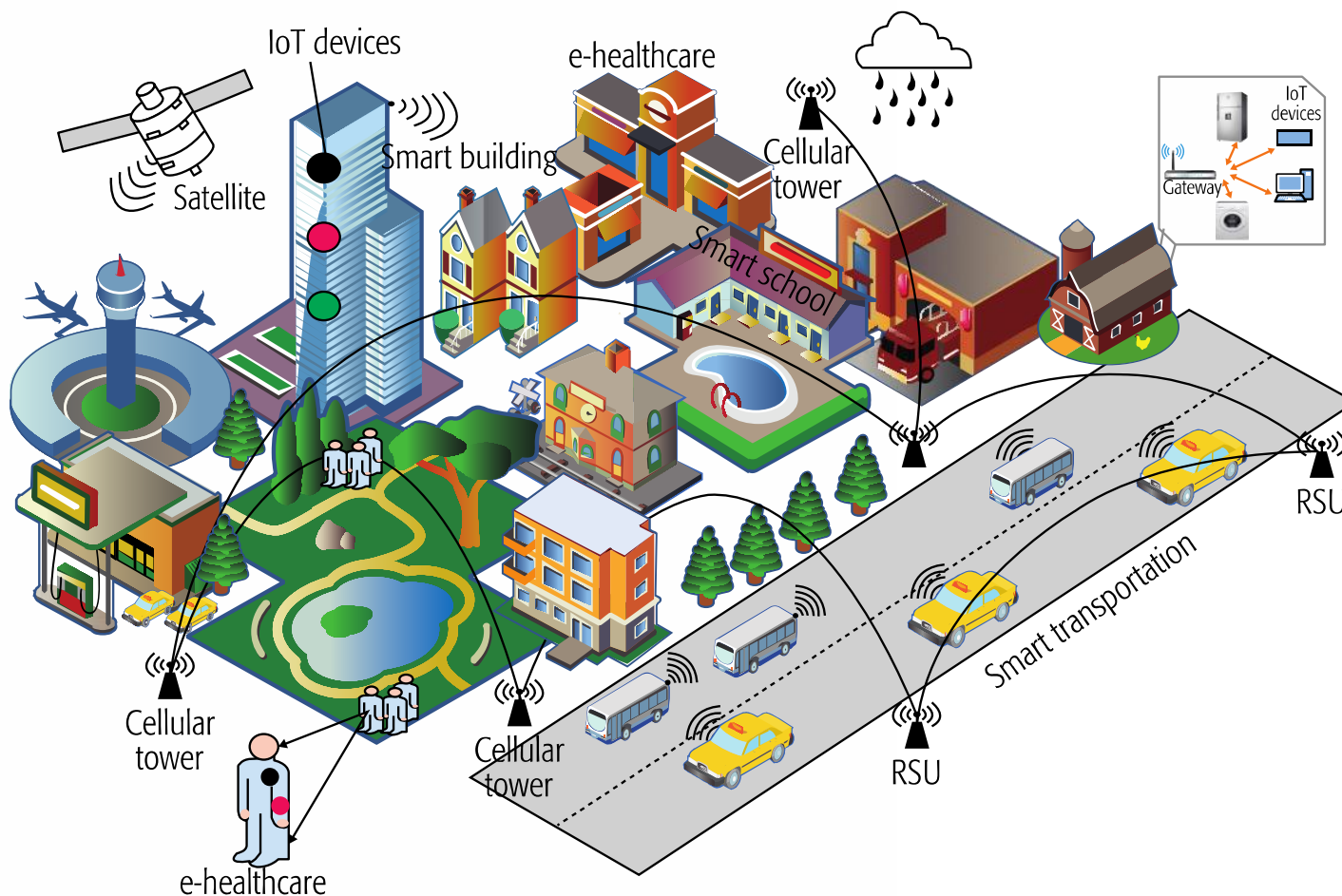
- Instrumenting buildings with IoT technologies may help in both **reducing the consumption of resources** associated to buildings (electricity, water) as well as in **improving the satisfaction level of humans** populating it
- A key role is played by **sensors**, which are used to both **monitor resource consumptions** as well as to **proactively detect current users' needs**
- Energy monitoring
- Home automation
- Home media services
- Home security
- Home comfort



Smart city

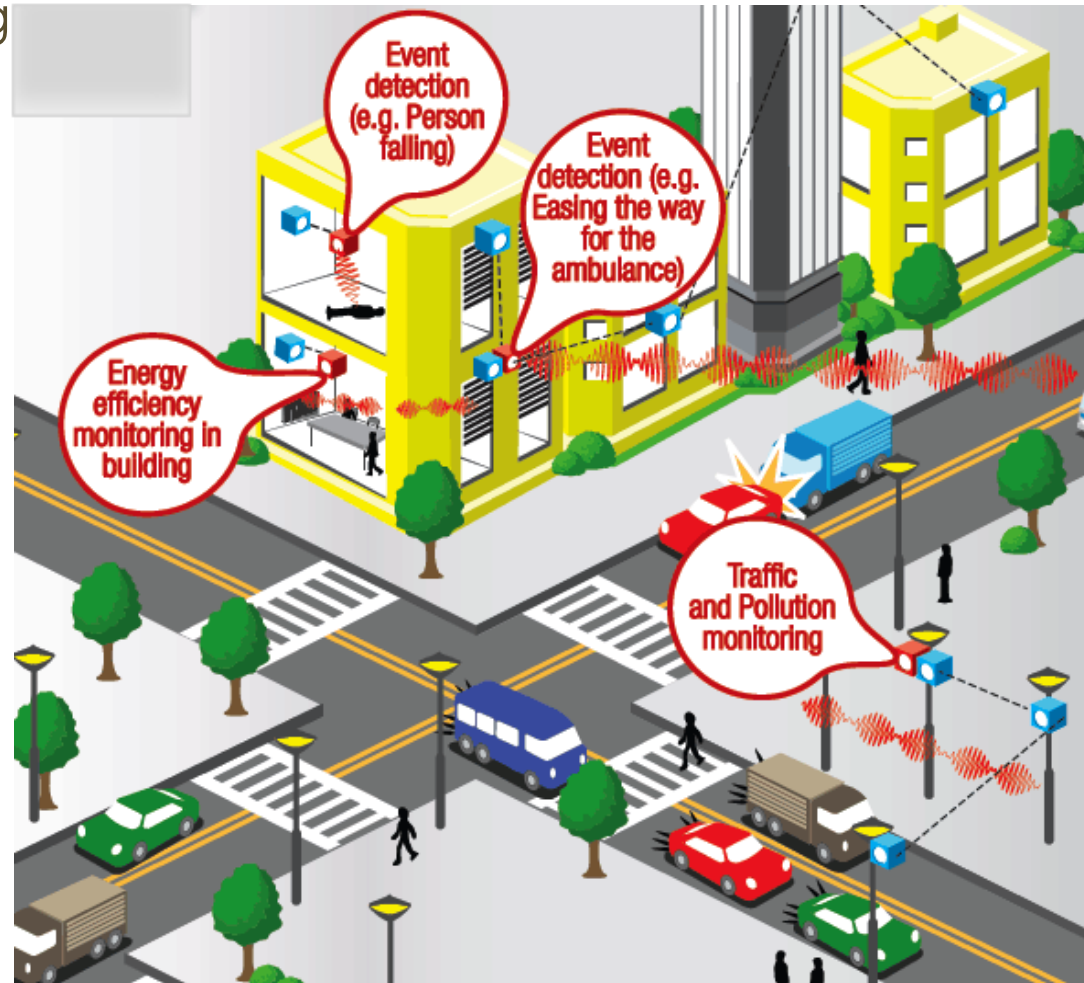


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Example: advanced traffic control system

- Car traffic monitoring in big cities or highways
- Traffic routing advice to avoid congestion
- Smart parking device system to provide drivers with automated parking advice
- Detecting violations
- Store information useful in case of accident





Dronet: networks of drones

Piloted drone

- Drones are used for many applications
- Aerial images
- Parcel delivery (amazon is experimenting a new service in the States)
- Most of applications **require the presence of human piloting**



From unmanned drones to autonomous drone networks

- **UNMANNED**
No human operator on board
(the drone may still be remotely piloted)
- **AUTONOMOUS**
Drones make decisions without continuous human control
- **MULTI-DRONE SYSTEM**
Drones need to communicate and coordinate
- **NETWORKING CHALLENGE**
How should drones exchange information, coordinate and route data autonomously?

Autonomy requires both coordination and communication.



Robophysical networks

- Robots need to sense the environment
- Robots need to communicate with nearby peers
- Robots need to coordinate their motion
- Communication and mobility continuously affect each other



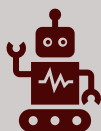
Communication shapes collective behavior — and collective behavior shapes the network.

From networking technologies to autonomy



We focus on autonomy mainly at the networking level

communication, channel access, routing
and, in some scenarios, mobility



How can a network make decisions autonomously?

What should it observe?
What decisions can it make?
How should it adapt?

How can autonomy be achieved?



There are different approaches to autonomous decision making



One possible answer is
Reinforcement Learning

Reinforcement Learning: the science of decision making

Overview of reinforcement learning (RL)



Reinforcement learning addresses a fundamental question:



How an intelligent agent can **learn** to make a **good sequence of decisions**

That's a single sentence that summarizes what reinforcement learning is.

Learning to make a good sequence of decisions (under uncertainty)

1. Learn to make good **sequence of decisions**

- How can an intelligent agent make not just one decision but a whole sequence of decisions

2. Learn to make **good** sequence of decisions

- Goodness: we have some utility measure over the decisions that are being made

3. **Learn** to make good sequence of decisions

- Learning
- The agent does not know in advance how its decisions are going to affect the world
- What decisions might be associated with good outcomes
- The agent has to acquire this information through experience

RL, Behavior & Intelligence

Why do we have a brain?

- Primitive creature which evolves as following during its lifetime
- Larval stage: primitive brain & eye, swims around, attaches to a rock
- Adult stage: sits, digests brain
- We need a brain to guide decisions
- **When all the decisions have been completed maybe we no longer need a brain**

Example from Yael Niv



Applications

- RL is applicable to a huge number of domains
- Incredible work on robotics: agents do grasping, fold clothes
- Robotics engineers at University of California at Berkeley have developed a robot that can fold towels and socks



Key aspects of RL



Optimization



Delayed consequences



Exploration



Generalization



RL: optimization



**Goal is to find an optimal way
to make decisions**

Yielding best outcomes



**Or at least a very good
strategy**

RL: delayed consequences



Decisions now can impact things much later...

ice cream example



Introduces two challenges

When **planning**: decisions involve reasoning about not just immediate benefit of a decision but also its longer term ramifications

When **learning**: temporal credit assignment is hard (what caused later high or low rewards?)



RL: exploration

- Learning about the world by making decisions
 - Learn to ride a bike by trying (and falling)
- Censored data (you only get to learn about what you try to do)
 - Only get a reward for decision made
 - Don't know what would have happened if we had taken a different decision
 - Each choice is made at a particular time

RL: generalization



Policy is mapping from past experience to action (defines the actions an agent should take based on its past experiences)



Why not just pre-program a policy?



We can have a huge number of states, thus we need generalization



Even if we run into a particular state we have never seen before our agent still know what to do (Generalization allows an RL agent to handle situations it has never encountered before by applying knowledge from similar past experiences.)



How do we proceed?



Explore the world



Use experience (rewards) to guide future decisions

Other issues



Where do rewards come from?

And what happens if we get it wrong?



Robustness/risk sensitivity

Robustness refers to the agent's ability to perform well in different, often unseen situations, without being too sensitive to slight changes in the environment

Risk sensitivity involves how the agent manages uncertainty, often opting for strategies that minimize risky outcomes



End of course goal

**Given a
networking
problem
(MAC or
routing)**

decide if it should be formulated as a
RL problem

if yes be able to define it formally (in
terms of state space, action space,
dynamics and reward model)

state what algorithm (from class) is
best suited to addressing it and
justify your answer



Next class

- We will begin an overview of wireless technologies
 - **RFID**
 - Sensor networks
 - IoT networks
 - Backscattering networks
 - Dronets